CS 423/523 – Fall 2025

ÖZYEĞİN ÜNİVERSİTESİ

Problem Set 4

Assigned: 15 December 2025
Due: 29 December 2025

What to Submit:

Create a folder named ps<*problem set number*>_<*YourLastName*> (for example, ps2_Ates), with the following structure and contents:

ps<*problem set number*>_<*problem number*>.py → The main script to run.

/input/ → Directory containing input images, videos or other data used

/output/ → Directory containing output image, videos or other data generated

ps<*problem set number*>_report.pdf → A PDF report file (Include figures, discussions, methods, etc., but do not include any code.) OR a Jupyter notebook that contains all codes and the content of the report together.

Zip your folder and submit on **LMS.** Submissions that do not obey the guideline above will **NOT** be evaluated.

Submissions without clear and detailed explanations of the codes provided and discussions of the results obtained will receive substantially lower grades!

Problem:

A folder with a set of images of a checkerboard pattern is provided. The size of a square in the pattern is 30mm. Using the images provided, do the following:

(a) Estimate the distortion parameters and undistort the images.
(b) Estimate the intrinsic matrix.
(c) Estimate the projection matrix for each image.
(d) Estimate the essential matrix between the first image and each of the remaining images.
(e) Estimate the rotation and translation between the reference image and each of the remaining images.

You may utilize the tutorial:

https://docs.opencv.org/4.x/dc/dbb/tutorial_py_calibration.html